Capstone 2 Project Report: The Longevity Shift

An Analysis of Ingredient Trends and Consumer Engagement in the Aesthetic Market

Student: Irina Kim

Mentor: Joe Xiao

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1. Problem Identification

The skincare and aesthetic industry is undergoing a paradigm shift. Consumers are moving away from "Aggressive Correction"—treatments like high-percentage acids and retinoids that require downtime—toward "Barrier Architecture" (Longevity). This category focuses on ingredients like Centella Asiatica, Ceramides, and Peptides that fortify the skin.

The Business Challenge: How can a brand predict "High Engagement" (Cult Status) for a product before launch? Using Sephora dataset metrics, we aim to identify the specific features—price, ingredient focus, and brand equity—that lead to a high "Loves Count."

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2. Methodology & Approach

I applied the **Data Science Method (DSM)** to move from raw metadata to a predictive classification model:

A. Data Wrangling & Engineering

- **Longevity Categorization:** I created a binary feature, `is_barrier`, by mining the `ingredients` column for keywords associated with the Longevity trend (Centella, Rice, Peptides, Ceramides).
- **Target Definition:** I defined "High Engagement" as any product with a `loves_count` exceeding the dataset median.
- **Cleaning:** Handled missing values in the `rating` and `price_usd` columns to ensure a robust training set of over 2,400 skincare products.

B. Preprocessing

- **Feature Scaling:** Applied `StandardScaler` to `price_usd` and `rating` to normalize the data for the Gradient Boosting algorithm.

- **Categorical Encoding:** Converted the Top 15 brands into dummy variables to measure the "Brand Multiplier" effect on engagement.

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3. Findings & Model Performance

I benchmarked three models (Logistic Regression, Random Forest, and Gradient Boosting). **Gradient Boosting** was selected as the champion model.

Metric	Gradient Boosting (Champion)	Random Forest	Logistic Regression
Accuracy	**67.7%**	66.8%	59.1%
Recall	**75.5%**	77.5%	53.4%
F1-Score	**0.70**	0.70	0.56

Key Insights from EDA:

- **The Satisfaction Floor:** Barrier Support products show more stable satisfaction (tighter rating distribution) than Aggressive Correction products, which suffer from "irritation-based" low ratings.
- **Feature Importance:** User **Rating (50.2%)** and **Price (29.6%)** are the most powerful predictors of whether a product will become a "Cult Favorite."

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4. Strategic Recommendations

Recommendation 1: The "Longevity" Price Point

Position new "Barrier Support" launches in the **$40–$80** price bracket. This segment shows the highest density of "Loves" and engagement, balancing perceived luxury with accessibility.

Recommendation 2: Clinical Synergy (Botox Support)

Develop and market "Post-Procedure Recovery" kits. Since barrier-focused products (Centella/Ceramides) have a higher satisfaction floor, they are the ideal daily companion for consumers undergoing professional clinical procedures like Botox.

Recommendation 3: Brand Equity Focus

The model highlights that brand names like *Drunk Elephant* and *Tatcha* act as significant engagement multipliers. Emerging Longevity brands (like *Skin1004*) should focus on "Aesthetic Packaging" and "Brand Storytelling" to reach the same engagement thresholds as luxury clinical players.

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5. Further Research

- **Ingredient Synergy:** Future research should use Natural Language Processing (NLP) on review text to identify *why* certain barrier ingredients correlate with high ratings (e.g., "glow," "no redness," "healing").
- **Temporal Trends:** Analyzing how the "Loves Count" grows over time would allow for a survival analysis of skincare products to predict shelf-life longevity.